

comparative - analysis -



How parameters affect on model

	increase	decrease
n_estimators	better accuracy more training time & memory usage	Fast training Less stable prediction
max_depth	Overfitting Captures complex patterns	Underfitting Simple model
min_samples_leaf	Reduce Overfitting May underfit	Overfitting More precise splits
random_state	This doesn't change performance, It makes the results repeatable. * Therefore always keep this fixed when comparing	
n_jobs	It will help to speed up training. There is no effect on accuracy.	

MAE - average absolute difference between actual vs predicted values

* $MAE \ll \text{target range [test]}$

* $\text{test_MAE} \approx \text{Train_MAE}$, if $\text{Test_MAE} \gg \text{Train_MAE}$ Overfitting

R^2 [coefficient of Determination] - How much variance is explained by the model

R^2

> 0.9 Excellent

$0.7 - 0.9$ Good

$0.5 - 0.7$ Acceptable

$0 - 0.5$ Weak

< 0 X

* $\text{Train } R^2 \uparrow$ but $\text{test } R^2 \downarrow \rightarrow \text{Overfitting}$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAPE

$< 5\%$ Excellent

$5\% - 10\%$ Very good

$10\% - 20\%$ Acceptable

$> 20\%$ Poor

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$R^2 = 1$ ✓

$R^2 = 0$ Predict mean

$R^2 < 0$ X

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i|} \times 100\%$$

target range = (0-800)

n_estimators = 100
max_depth = 15
min_sample_leaf = 5
random_state = 42
n_jobs = -1

Test_data

Rev.center 1	MAE	R ²	MAPE	Rev.center 2	MAE	R ²	MAPE
Breakfast	645	0.278	∞	Breakfast	263	0.170	∞
Lunch	393	0.239	∞	Lunch	488	0.188	∞
Dinner	974	0.377	44.7%	Dinner	341	0.193	∞
Rev.center 3	MAE	R ²	MAPE	Rev.center 4	MAE	R ²	MAPE
Breakfast	250	0.008	∞	Breakfast	0	1	0
Lunch	847	0.316	56.5%	Lunch	0	1	0
Dinner	3844	0.403	34.1%	Dinner	0	0	∞
Rev.center 5	MAE	R ²	MAPE	Rev.center 6	MAE	R ²	MAPE
Breakfast	2548	0.592	16.6%	Breakfast	195	0.294	0%
Lunch	1721	0.086	∞%	Lunch	423	0.186	∞%
Dinner	6391	0.285	79.1%	Dinner	1825	0.193	∞%
Rev.center 7	MAE	R ²	MAPE	Rev.center 8	MAE	R ²	MAPE
Breakfast	22	0.385	∞%	Breakfast	2260	0.065	0%
Lunch	37	0.159	∞%	Lunch	8959	0.111	∞%
Dinner	79	0.189	∞%	Dinner	18.018	0.097	∞%
Rev.center 9	MAE	R ²	MAPE				
Breakfast	242	-0.22	∞%				
Lunch	146	0.117	∞%				
Dinner	84	0.123	∞%				

- conclusion -

MAE(test) should be << 800 ∴ Rev center 4, 7, 9
not suitable for Rev center 8, 5

suitable for some meal periods in other revenue center

Training data

Rev. center 1	MAE	R ²	MAPE	Rev. center 2	MAE	R ²	MAPE
Breakfast	276	0.706	∞%	Breakfast	140	0.619	∞%
Lunch	240	0.682	∞%	Lunch	166	0.608	∞%
Dinner	477	0.83	22.1%	Dinner	217	0.498	∞%
Rev. center 3	MAE	R ²	MAPE	Rev. center 4	MAE	R ²	MAPE
Breakfast	160	0.599	∞%	Breakfast	0	1	0%
Lunch	467	0.628	36.8%	Lunch	0	1	0%
Dinner	1523	0.863	14.7%	Dinner	0	0.111	∞%
Rev. center 5	MAE	R ²	MAPE	Rev. center 6	MAE	R ²	MAPE
Breakfast	976	0.939	6.2%	Breakfast	94	0.592	0%
Lunch	785	0.706	∞%	Lunch	165	0.66	∞%
Dinner	2137	0.698	56.7%	Dinner	615	0.311	∞%
Rev. center 7	MAE	R ²	MAPE	Rev. center 8	MAE	R ²	MAPE
Breakfast	26	0.529	∞	Breakfast	804	0.306	0%
Lunch	20	0.523	∞	Lunch	4934	0.626	∞%
Dinner	32	0.619	∞	Dinner	8100	0.595	∞%
Rev. center 9	MAE	R ²	MAPE				
Breakfast	91	0.716	∞%				
Lunch	96	0.582	∞%				
Dinner	53	0.466	∞%				

* This model is very suitable for rev. center 4 (both MAE equals and small), rev. center 9 (small and relatively equal) and 7 (small and relatively equal),
2.

target range 0-500

n_estimators = 200
 max_depth = 20
 min_sample_leaf = 10
 random_state = 60
 n_jobs = -1

test data

Rev. center 1	MAE	R ²	MAPE
Breakfast	681	0.244	∞ %
Lunch	387	0.36	∞ %
Dinner	929	0.445	42.8 %

Rev. center 2	MAE	R ²	MAPE
Breakfast	259	0.215	∞ %
Lunch	459	0.172	∞ %
Dinner	321	0.257	∞ %

Rev. center 3	MAE	R ²	MAPE
Breakfast	249	0.025	∞ %
Lunch	844	0.319	56.8 %
Dinner	3674	0.432	33.0 %

Rev. center 4	MAE	R ²	MAPE
Breakfast	0	1	0 %
Lunch	0	1	0 %
Dinner	0	0	∞ %

Rev. center 5	MAE	R ²	MAPE
Breakfast	2615	0.516	16.9 %
Lunch	1887	-0.031	∞ %
Dinner	6497	0.24	80.7 %

Rev. center 6	MAE	R ²	MAPE
Breakfast	196	0.214	∞ %
Lunch	408	0.219	∞ %
Dinner	1663	0.213	∞ %

Rev. center 7	MAE	R ²	MAPE
Breakfast	21	0.371	∞ %
Lunch	39	0.14	∞ %
Dinner	77	0.162	∞ %

Rev. center 8	MAE	R ²	MAPE
Breakfast	1981	0.167	∞ %
Lunch	9460	0.061	∞ %
Dinner	17941	0.108	∞ %

Rev. center 9	MAE	R ²	MAPE
Breakfast	229	-0.036	∞ %
Lunch	149	0.085	∞ %
Dinner	81	0.143	∞ %

suitable for: rev center 4, 7, 9, 2 & some meal periods for other centers.

training data

Rev. center 1	MAE	R ²	MAPE	Rev. center 2	MAE	R ²	MAPE
Breakfast	336	0.598	∞%	Breakfast	168	0.482	∞%
Lunch	284	0.562	∞%	Lunch	200	0.459	∞%
Dinner	593	0.738	27.1%	Dinner	253	0.392	∞%
Rev. center 3	MAE	R ²	MAPE	Rev. center 4	MAE	R ²	MAPE
Breakfast	194	0.429	∞%	Breakfast	0	1	0%
Lunch	553	0.498	43.9%	Lunch	0	1	0%
Dinner	1823	0.811	17.8%	Dinner	0	0	∞%
Rev. center 5	MAE	R ²	MAPE	Rev. center 6	MAE	R ²	MAPE
Breakfast	1174	0.913	7.4%	Breakfast	112	0.437	∞%
Lunch	995	0.553	∞%	Lunch	198	0.533	∞%
Dinner	2581	0.585	67.2%	Dinner	697	0.189	∞%
Rev. center 7	MAE	R ²	MAPE	Rev. center 8	MAE	R ²	MAPE
Breakfast	31	0.37	∞%	Breakfast	850	0.22	0%
Lunch	24	0.378	∞%	Lunch	5915	0.468	∞%
Dinner	38	0.475	∞%	Dinner	9584	0.457	∞%
Rev. center 9	MAE	R ²	MAPE				
Breakfast	107	0.616	∞%				
Lunch	114	0.443	∞%				
Dinner	59	0.333	∞%				

This method is very suitable for 4, 7, 9, 2

target range 0-800

n_estimators = 300
 max_depth = 30
 min_sample_leaf = 20
 random_state = 80
 n_jobs = -1

test data

Rev. center 1	MAE	R ²	MAPE	Rev. center 2	MAE	R ²	MAPE
Breakfast	687	0.229	∞%	Breakfast	254	0.228	∞%
Lunch	383	0.404	∞%	Lunch	458	0.132	∞%
Dinner	915	0.462	41.9%	Dinner	363	0.159	∞%
Rev. center 3	MAE	R ²	MAPE	Rev. center 4	MAE	R ²	MAPE
Breakfast	256	0.011	∞%	Breakfast	0	1	0
Lunch	816	0.368	54.5%	Lunch	0	1	0
Dinner	3655	0.392	31.9%	Dinner	0	0	∞%
Rev. center 5	MAE	R ²	MAPE	Rev. center 6	MAE	R ²	MAPE
Breakfast	2632	0.519	16.8%	Breakfast	196	0.184	∞%
Lunch	1819	0.045	∞%	Lunch	400	0.244	∞%
Dinner	6517	0.238	85.3%	Dinner	1501	0.086	∞%
Rev. center 7	MAE	R ²	MAPE	Rev. center 8	MAE	R ²	MAPE
Breakfast	21	0.363	∞%	Breakfast	1859	0.137	∞%
Lunch	38	0.15	∞%	Lunch	8521	0.126	∞%
Dinner	74	0.161	∞%	Dinner	17459	0.141	∞%
Rev. center 9	MAE	R ²	MAPE				
Breakfast	211	0.184	∞%				
Lunch	149	0.098	∞%				
Dinner	80	0.15	∞%				

suitable for - rev center 4, 7, 9, 2 & some meal periods for other centers.

training data

Rev. center 1	MAE	R ²	MAPE	Rev. center 2	MAE	R ²	MAPE
Breakfast	373	0.534	∞%	Breakfast	188	0.368	∞%
Lunch	313	0.469	∞%	Lunch	222	0.354	∞%
Dinner	679	0.66	30.7%	Dinner	285	0.292	∞%
Rev. center 3	MAE	R ²	MAPE	Rev. center 4	MAE	R ²	MAPE
Breakfast	216	0.299	∞%	Breakfast	0	1	0
Lunch	614	0.394	49.0%	Lunch	0	1	0
Dinner	2050	0.751	19.9%	Dinner	0	0	∞%
Rev. center 5	MAE	R ²	MAPE	Rev. center 6	MAE	R ²	MAPE
Breakfast	1342	0.887	8.4%	Breakfast	123	0.33	∞%
Lunch	1187	0.421	∞%	Lunch	219	0.444	∞%
Dinner	2929	0.488	76.8%	Dinner	717	0.139	∞%
Rev. center 7	MAE	R ²	MAPE	Rev. center 8	MAE	R ²	MAPE
Breakfast	33	0.284	∞%	Breakfast	883	0.173	∞%
Lunch	26	0.276	∞%	Lunch	6443	0.349	∞%
Dinner	42	0.355	∞%	Dinner	10722	0.331	∞%
Rev. center 9	MAE	R ²	MAPE				
Breakfast	120	0.496	∞%				
Lunch	127	0.327	∞%				
Dinner	63	0.248	∞%				

This method is very suitable for 4, 7, 9, 3, 2

Final conclusion

Random Forest method is very suitable for predict future revenue on revenue center 4, 7, 9